

Московский физико-технический институт
Высшая школа программной инженерии

Базы данных
3 семестр, 2026/2027 уч. год

Лекция 1

Основы SQL

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1 Типы баз данных

- **Сетевые и иерархические** — данные организованы в виде иерархии/сети.
- **Реляционные** — данные представлены в виде таблиц, связанных между собой.
- **Big Data, NoSQL** — данные хранятся в неструктурированном виде на распределённом кластере серверов.

2 История

- **1970** — реляционная модель данных. Эдгар Кодд разработал реляционную модель данных.
- **1974** — язык SEQUEL. Реймонд Бойс и Дональд Чамберлин в IBM создали язык запросов к реляционной бд SEQUEL (Structured English Query Language).
- **1979** — выход систем управления бд с поддержкой SEQUEL(sql).
- **1986** — стандартизация SQL. Последняя версия стандарта — SQL:2023 (с ревизией от 2026 года).

3 СУБД

Популярные реляционные СУБД: Oracle Database, Microsoft SQL Server, IBM DB2, MySQL, MariaDB, PostgreSQL, SQLite.

Популярность СУБД основана на опросе от ([опросе от stack overflow в 2025](#)): PostgreSQL — 55,6 %, MySQL — 40,5 %, SQLite — 37,5 %, Microsoft SQL Server — 30,1 %, Redis — 28 %, MongoDB — 24 %, MariaDB — 22,5 %, Elasticsearch — 16,7 %, Oracle — 10,6 %.

Еще SQL используют в нереляционных базах данных: Apache Hive, Apache Spark SQL, Apache Drill.

3.1 Почему SQL популярен

- Строгая математическая основа: реляционная модель данных, реляционная алгебра, реляционное исчисление.
- Стандарты: ANSI SQL-86 — SQL:2023.
- Большое количество людей знают SQL: многие компании, которые делают системы управления базами данных реализуют поддержку SQL для расширения количества пользователей.

4 Таблица супергероев

далее примеры на основе [разных героев комиксов и вселенных](#),

id	name	align	eye	hair	gender	appearances	year	universe
1	Spider-Man (Peter Parker)	Good Characters	Hazel Eyes	Brown Hair	Male Characters	4043	1962	marvel
2	Captain America (Steven Rogers)	Good Characters	Blue Eyes	White Hair	Male Characters	3360	1941	marvel
4854	Batman (Bruce Wayne)	Good Characters	Blue Eyes	Black Hair	Male Characters	3093	1939	dc
3	Wolverine (James \"Logan\" Howlett)	Neutral Characters	Blue Eyes	Black Hair	Male Characters	3061	1974	marvel
4	Iron Man (Anthony \"Tony\" Stark)	Good Characters	Blue Eyes	Black Hair	Male Characters	2961	1963	marvel
4855	Superman (Clark Kent)	Good Characters	Blue Eyes	Black Hair	Male Characters	2496	1986	dc
5	Thor (Thor Odinson)	Good Characters	Blue Eyes	Blond Hair	Male Characters	2258	1950	marvel
6	Benjamin Grimm (Earth-616)	Good Characters	Blue Eyes	No Hair	Male Characters	2255	1961	marvel

5 Оператор SELECT

```
SELECT * FROM superheroes
```

id	name	align	eye	hair	gender	appearances	year	universe
1	Spider-Man (Peter Parker)	Good Characters	Hazel Eyes	Brown Hair	Male Characters	4043	1962	marvel
2	Captain America (Steven Rogers)	Good Characters	Blue Eyes	White Hair	Male Characters	3360	1941	marvel
4854	Batman (Bruce Wayne)	Good Characters	Blue Eyes	Black Hair	Male Characters	3093	1939	dc
3	Wolverine (James \"Logan\" Howlett)	Neutral Characters	Blue Eyes	Black Hair	Male Characters	3061	1974	marvel
4	Iron Man (Anthony \"Tony\" Stark)	Good Characters	Blue Eyes	Black Hair	Male Characters	2961	1963	marvel
4855	Superman (Clark Kent)	Good Characters	Blue Eyes	Black Hair	Male Characters	2496	1986	dc
5	Thor (Thor Odinson)	Good Characters	Blue Eyes	Blond Hair	Male Characters	2258	1950	marvel
6	Benjamin Grimm (Earth-616)	Good Characters	Blue Eyes	No Hair	Male Characters	2255	1961	marvel
			...					

Глобально мы получим вывод как сверху, выбираем все столбцы из таблицы `superheroes`

5.1 Выбор столбцов в SELECT

```
SELECT name, appearances
FROM superheroes
```

name	appearances
Spider-Man (Peter Parker)	4043
Captain America (Steven Rogers)	3360
Wolverine (James \"Logan\" Howlett)	3061
Iron Man (Anthony \"Tony\" Stark)	2961
Thor (Thor Odinson)	2258
Benjamin Grimm (Earth-616)	2255
Reed Richards (Earth-616)	2072
Hulk (Robert Bruce Banner)	2017
...	

5.2 Псевдонимы столбцов в SELECT

```
SELECT name AS hero_name,
       appearances
FROM superheroes
```

hero_name	appearances
Spider-Man (Peter Parker)	4043
Captain America (Steven Rogers)	3360
Wolverine (James \"Logan\" Howlett)	3061
Iron Man (Anthony \"Tony\" Stark)	2961
Thor (Thor Odinson)	2258
Benjamin Grimm (Earth-616)	2255
Reed Richards (Earth-616)	2072
Hulk (Robert Bruce Banner)	2017
...	

формально AS можно опустить и писать так:

```
SELECT name hero_name,  
       appearances  
FROM superheroes
```

Но тогда читаемость падает, но это выбор каждого

5.3 Выбор уникальных значений

```
SELECT DISTINCT align  
FROM superheroes
```

align
Good Characters
Neutral Characters
Reformed Criminals
Bad Characters

```
SELECT DISTINCT eye  
FROM superheroes
```

eye
Black Eyeballs
One Eye
Purple Eyes
Orange Eyes
...
(24 строки)

Результат: Black Eyeballs, One Eye, Purple Eyes, Orange Eyes, ... (24 строки).

5.4 Ограничение количества строк

```
SELECT DISTINCT hair  
FROM superheroes  
LIMIT 10
```

hair
Strawberry Blond Hair
Violet Hair
Green Hair
Bald
Silver Hair
Light Brown Hair
Red Hair
White Hair
Purple Hair
Reddish Blond Hair

6 Выбор строк: WHERE

```
SELECT *
FROM superheroes
WHERE gender = 'Female Characters'
```

id	name	align	eye	hair	gender	appearances	year	universe
12	Susan Storm (Earth-616)	Good Characters	Blue Eyes	Blond Hair	Female Characters	1713	1961	marvel
13	Ororo Munroe (Earth-616)	Good Characters	Blue Eyes	White Hair	Female Characters	1512	1975	marvel
17	Mary Jane Watson (Earth-616)	Good Characters	Green Eyes	Red Hair	Female Characters	1304	1965	marvel
24	Wanda Maximoff (Earth-616)	Good Characters	Green Eyes	Brown Hair	Female Characters	1161	1964	marvel
26	Janet van Dyne (Earth-616)	Good Characters	Blue Eyes	Auburn Hair	Female Characters	1120	1963	marvel
27	Jean Grey (Earth-616)	Good Characters	Green Eyes	Red Hair	Female Characters	1107	1963	marvel
28	Natalia Romanova (Earth-616)	Good Characters	Brown Eyes	Red Hair	Female Characters	1050	1964	marvel
31	May Reilly (Earth-616)	Good Characters	Blue Eyes	Grey Hair	Female Characters	1000	1962	marvel
			...					

6.1 Операторы сравнения

Оператор	Назначение
=	Равно
<>, !=	Неравно
>	Больше
>=	Больше или равно
<	Меньше
<=	Меньше или равно
between	Значение находится в указанном диапазоне
in	Значение входит в список
like	Проверка строки на соответствие шаблону

6.2 Использование BETWEEN в WHERE

```
SELECT *
FROM superheroes
WHERE year BETWEEN 2000 AND 2005
```

id	name	align	eye	hair	gender	appearances	year	universe
111	Maria Hill (Earth-616)	Good Characters	Brown Eyes	Black Hair	Female Characters	325	2005	marvel
140	Laura Kinney (Earth-616)	Good Characters	Green Eyes	Black Hair	Female Characters	265	2004	marvel
164	Robert Reynolds (Earth-616)	Neutral Characters	Blue Eyes	Blond Hair	Male Characters	240	2000	marvel
166	Santo Vaccarro (Earth-616)	Good Characters	White Eyes	No Hair	Male Characters	238	2003	marvel
174	Megan Gwynn (Earth-616)	Good Characters	Black Eyeballs	Pink Hair	Female Characters	229	2004	marvel
211	Victor Borkowski (Earth-616)	Good Characters	Brown Eyes	Black Hair	Male Characters	184	2003	marvel
217	Julian Keller (Earth-616)	Good Characters	Blue Eyes	Black Hair	Male Characters	179	2003	marvel
219	Celeste Cuckoo (Earth-616)	Good Characters	Blue Eyes	Blond Hair	Female Characters	177	2001	marvel
			...					

6.3 Использование IN в WHERE

```
SELECT *
FROM superheroes
WHERE hair IN ('Strawberry Blond Hair', 'Red Hair', 'Auburn Hair')
```

id	name	align	eye	hair	gender	appearances	year	universe
15	Matthew Murdock (Earth-616)	Good Characters	Blue Eyes	Red Hair	Male Characters	1338	1338	marvel
17	Mary Jane Watson (Earth-616)	Good Characters	Green Eyes	Red Hair	Female Characters	1304	1965	marvel
26	Janet van Dyne (Earth-616)	Good Characters	Blue Eyes	Auburn Hair	Female Characters	1120	1963	marvel
27	Jean Grey (Earth-616)	Good Characters	Green Eyes	Red Hair	Female Characters	1107	1963	marvel
28	Natalia Romanova (Earth-616)	Good Characters	Brown Eyes	Red Hair	Female Characters	1050	1964	marvel
46	Norman Osborn (Earth-616)	Bad Characters	Blue Eyes	Auburn Hair	Male Characters	692	1964	marvel
47	Eugene Thompson (Earth-616)	Good Characters	Blue Eyes	Strawberry Blond Hair	Male Characters	689	1962	marvel
58	Patricia Walker (Earth-616)	Good Characters	Blue Eyes	Red Hair	Female Characters	550	1944	marvel
...								

6.4 Использование LIKE в WHERE

Специальные символы в шаблонах LIKE:

- % — любое количество любых символов (включая 0);
- _ — ровно один любой символ.

```
SELECT *
FROM superheroes
WHERE hair LIKE '%Blond%'
```

id	name	align	eye	hair	gender	appearances	year	universe
5	Thor (Thor Odinson)	Good Characters	Blue Eyes	Blond Hair	Male Characters	2258	1950	marvel
10	Jonathan Storm (Earth-616)	Good Characters	Blue Eyes	Blond Hair	Male Characters	1934	1961	marvel
12	Susan Storm (Earth-616)	Good Characters	Blue Eyes	Blond Hair	Female Characters	1713	1961	marvel
14	Clinton Barton (Earth-616)	Good Characters	Blue Eyes	Blond Hair	Male Characters	1394	1964	marvel
47	Eugene Thompson (Earth-616)	Good Characters	Blue Eyes	Strawberry Blond Hair	Male Characters	689	1962	marvel
52	Daniel Rand (Earth-616)	Good Characters	Blue Eyes	Blond Hair	Male Characters	618	1974	marvel
66	Crystallia Amaquelin (Earth-616)	Good Characters	Green Eyes	Strawberry Blond Hair	Female Characters	512	1965	marvel
79	Johnathon Blaze (Earth-616)	Good Characters	Blue Eyes	Reddish Blond Hair	Male Characters	426	1972	marvel
...								

6.5 Логические операции

Операция	Назначение
AND	Логическое И
OR	Логическое ИЛИ
NOT	Логическое НЕ

6.5.1 Использование AND в WHERE

```
SELECT *
FROM superheroes
WHERE gender = 'Female Characters'
AND
      align = 'Bad Characters'
```

id	name	align	eye	hair	gender	appearances	year	universe
99	Raven Darkholme (Earth-616)	Bad Characters	Yellow Eyes	Red Hair	Female Characters	371	1978	marvel
157	Amora (Earth-616)	Bad Characters	Green Eyes	Blond Hair	Female Characters	247	1964	marvel
221	Ophelia Sarkissian (Earth-616)	Bad Characters	Green Eyes	Black Hair	Female Characters	175	1969	marvel
229	Hela (Earth-616)	Bad Characters	Green Eyes	Black Hair	Female Characters	170	1964	marvel
253	Veranke (Earth-616)	Bad Characters	Green Eyes	Green Hair	Female Characters	156	2005	marvel
264	Giuletta Nefaria (Earth-616)	Bad Characters	Grey Eyes	Black Hair	Female Characters	149	1968	marvel
272	Madelyne Pryor (Earth-616)	Bad Characters	Green Eyes	Red Hair	Female Characters	146	1983	marvel
356	Mary Walker (Earth-616)	Bad Characters	Brown Eyes	Brown Hair	Female Characters	111	1988	marvel
			...					

6.5.2 Использование OR в WHERE

```
SELECT *
FROM superheroes
WHERE hair = 'Red Hair'
OR hair = 'Strawberry Blond Hair'
OR hair = 'Auburn Hair'
```

id	name	align	eye	hair	gender	appearances	year	universe
15	Matthew Murdock (Earth-616)	Good Characters	Blue Eyes	Red Hair	Male Characters	1338	1338	marvel
17	Mary Jane Watson (Earth-616)	Good Characters	Green Eyes	Red Hair	Female Characters	1304	1965	marvel
26	Janet van Dyne (Earth-616)	Good Characters	Blue Eyes	Auburn Hair	Female Characters	1120	1963	marvel
27	Jean Grey (Earth-616)	Good Characters	Green Eyes	Red Hair	Female Characters	1107	1963	marvel
28	Natalia Romanova (Earth-616)	Good Characters	Brown Eyes	Red Hair	Female Characters	1050	1964	marvel
46	Norman Osborn (Earth-616)	Bad Characters	Blue Eyes	Auburn Hair	Male Characters	692	1964	marvel
47	Eugene Thompson (Earth-616)	Good Characters	Blue Eyes	Strawberry Blond Hair	Male Characters	689	1962	marvel
58	Patricia Walker (Earth-616)	Good Characters	Blue Eyes	Red Hair	Female Characters	550	1944	marvel
			...					

6.5.3 Использование NOT в WHERE

```
SELECT *
FROM superheroes
WHERE hair NOT IN ('Blond Hair', 'Black Hair', 'Brown Hair', 'Red Hair')
```

id	name	align	eye	hair	gender	appearances	year	universe
2	Captain America (Steven Rogers)	Good Characters	Blue Eyes	White Hair	Male Characters	3360	1941	marvel
6	Benjamin Grimm (Earth-616)	Good Characters	Blue Eyes	No Hair	Male Characters	2255	1961	marvel
11	Henry McCoy (Earth-616)	Neutral Characters	Blue Eyes	Blue Hair	Male Characters	1825	1974	marvel
13	Ororo Munroe (Earth-616)	Good Characters	Blue Eyes	White Hair	Female Characters	1512	1975	marvel
21	Charles Xavier (Earth-616)	Good Characters	Blue Eyes	Bald	Male Characters	1233	1963	marvel
26	Janet van Dyne (Earth-616)	Good Characters	Blue Eyes	Auburn Hair	Female Characters	1120	1963	marvel
29	Kurt Wagner (Earth-616)	Good Characters	Yellow Eyes	Blue Hair	Male Characters	1047	1975	marvel
30	Vision (Earth-616)	Good Characters	Gold Eyes	No Hair	Female Characters	1007	1968	marvel
			...					

7 Сортировка: ORDER BY

```
SELECT *
FROM superheroes
ORDER BY year
```

id	name	align	eye	hair	gender	appearances	year	universe
5352	Arthur Pendragon (New Earth)	Good Characters	Brown Eyes	Brown Hair	Male Characters	41	1936	dc
5959	Lady of the Lake (New Earth)	Good Characters	Blue Eyes	Blue Hair	Female Characters	13	1936	dc
5094	Merlin (New Earth)	Neutral Characters	Black Eyes	White Hair	Male Characters	92	1936	dc
5143	Cyril Saunders (New Earth)	Good Characters	Hazel Eyes	White Hair	Male Characters	79	1937	dc
4941	Samuel Bradley (New Earth)	Good Characters	Grey Eyes	Grey Hair	Male Characters	213	1937	dc
6087	Rose Psychic (New Earth)	Good Characters	Black Eyes	Black Hair	Female Characters	11	1937	dc
4913	John Zatara (New Earth)	Good Characters	Blue Eyes	Black Hair	Male Characters	256	1938	dc
4868	Lois Lane (New Earth)	Good Characters	Blue Eyes	Black Hair	Female Characters	934	1938	dc
			...					

7.1 Порядок сортировки

Порядок сортировки в ORDER BY:

- ASC (ascending) — сортировка по возрастанию;
- DESC (descending) — сортировка по убыванию.

```
SELECT *
FROM superheroes
ORDER BY appearances DESC
```

id	name	align	eye	hair	gender	appearances	year	universe
1	Spider-Man (Peter Parker)	Good Characters	Hazel Eyes	Brown Hair	Male Characters	4043	1962	marvel
2	Captain America (Steven Rogers)	Good Characters	Blue Eyes	White Hair	Male Characters	3360	1941	marvel
4854	Batman (Bruce Wayne)	Good Characters	Blue Eyes	Black Hair	Male Characters	3093	1939	dc
3	Wolverine (James \"Logan\" Howlett)	Neutral Characters	Blue Eyes	Black Hair	Male Characters	3061	1974	marvel
4	Iron Man (Anthony \"Tony\" Stark)	Good Characters	Blue Eyes	Black Hair	Male Characters	2961	1963	marvel
4855	Superman (Clark Kent)	Good Characters	Blue Eyes	Black Hair	Male Characters	2496	1986	dc
5	Thor (Thor Odinson)	Good Characters	Blue Eyes	Blond Hair	Male Characters	2258	1950	marvel
6	Benjamin Grimm (Earth-616)	Good Characters	Blue Eyes	No Hair	Male Characters	2255	1961	marvel
...								

7.2 Сортировка и ограничение количества строк

```
SELECT *
FROM superheroes
WHERE align = 'Bad Characters'
      AND gender = 'Female Characters'
ORDER BY appearances DESC
LIMIT 5
```

id	name	align	eye	hair	gender	appearances	year	universe
99	Raven Darkholme (Earth-616)	Bad Characters	Yellow Eyes	Red Hair	Female Characters	371	1978	marvel
157	Amora (Earth-616)	Bad Characters	Green Eyes	Blond Hair	Female Characters	247	1964	marvel
4956	Talia al Ghul (New Earth)	Bad Characters	Brown Eyes	Brown Hair	Female Characters	177	1971	dc
221	Ophelia Sarkissian (Earth-616)	Bad Characters	Green Eyes	Black Hair	Female Characters	175	1969	marvel
229	Hela (Earth-616)	Bad Characters	Green Eyes	Black Hair	Female Characters	170	1964	marvel

7.3 Сортировка по нескольким столбцам

```
SELECT *
FROM superheroes
ORDER BY year, appearances
```

id	name	align	eye	hair	gender	appearances	year	universe
5959	Lady of the Lake (New Earth)	Good Characters	Blue Eyes	Blue Hair	Female Characters	13	1936	dc
5352	Arthur Pendragon (New Earth)	Good Characters	Brown Eyes	Brown Hair	Male Characters	41	1936	dc
5094	Merlin (New Earth)	Neutral Characters	Black Eyes	White Hair	Male Characters	92	1936	dc
6087	Rose Psychic (New Earth)	Good Characters	Black Eyes	Black Hair	Female Characters	11	1937	dc
5143	Cyril Saunders (New Earth)	Good Characters	Hazel Eyes	White Hair	Male Characters	79	1937	dc
4941	Samuel Bradley (New Earth)	Good Characters	Grey Eyes	Grey Hair	Male Characters	213	1937	dc
5069	Wing How (New Earth)	Good Characters	Brown Eyes	Black Hair	Male Characters	103	1938	dc
			...					

8 SQL — декларативный язык

В джаве и с++ мы привыкли описывать как получить результат, а тут мы описываем какой хотим увидеть результат. В этом и основное отличие sql.

9 Создание и изменение таблиц

```
CREATE TABLE superheroes(
  id INT,
  name VARCHAR(100),
  align VARCHAR(30),
  eye VARCHAR(30),
  hair VARCHAR(30),
  gender VARCHAR(30),
  appearances INT,
  year INT,
  universe VARCHAR(10)
)
```

9.1 Типы данных в SQL

Тип данных	Назначение
CHARACTER(n) (CHAR(n))	Строка фиксированной длины n
CHARACTER VARYING(n) (VARCHAR(n))	Строка переменной длины, максимальная длина n
BOOLEAN	Логический тип данных
INTEGER (INT)	Целое число
NUMERIC(p, s)	Действительное число (p — количество значащих цифр, s — количество цифр после запятой). Хранится точно
REAL	Действительное число одинарной точности, формат IEEE 754
DOUBLE PRECISION	Действительное число двойной точности, формат IEEE 754
DATE	Дата
TIMESTAMP	Дата и время

Иногда деньги хранятся в виде целого числа, представляющего количество центов, чтобы избежать проблем с точностью. А иногда используют DECIMAL или NUMERIC с фиксированной точностью. Не стоит хранить в даблах или флоатах, так как из-за специфики хранения этих чисел, можно в какой-то момент получить у пользователь на копейку больше или меньше, чем он должен был получить.

9.2 Первичный ключ

```
CREATE TABLE superheroes(  
    id INT PRIMARY KEY,  
    ...  
)
```

Заполнение первичного ключа, чтобы он сам проставлялся в таблице при добавлении новых строк:

```
CREATE TABLE superheroes(  
    id INT PRIMARY KEY GENERATED BY DEFAULT AS IDENTITY,  
    ...  
)
```

```
CREATE TABLE superheroes(  
    id SERIAL PRIMARY KEY,  
    ...  
)
```

9.3 Удаление таблицы

```
DROP TABLE superheroes
```

9.4 Изменение таблиц

```
ALTER TABLE superheroes ADD COLUMN alive BOOLEAN;
```

```
ALTER TABLE superheroes  
    ADD COLUMN first_appearance TIMESTAMP;
```

```
ALTER TABLE superheroes DROP COLUMN year;
```

```
ALTER TABLE superheroes RENAME COLUMN name TO hero_name;
```

```
ALTER TABLE superheroes RENAME TO comic_characters;
```



Каринок на эту часть нет, держите Уолтер Вайта